

PED – 6 THE CHILD AND ADOLESCENT LEARNERS AND LEARNING PRINCIPLES

Learner-Centered Psychological Principles (LCP)

- Focus on psychological factors internal to the learner; acknowledge external environment that interacts with internal factors
- Deal holistically with learners in the context of real-world learning situations
- Intended to apply to all learners
 - Children
 - Teachers
 - Administrators
 - parents
 - community members

Cognitive and Metacognitive Factors

1. Nature of the learning process

- The learning of complex subject matter is most effective when it is an intentional process of constructing meaning from information and experience.

2. Goals of the learning process

- The successful learner, over time and with support and instructional guidance, can create meaningful, coherent representations of knowledge.

3. Construction of knowledge

- The successful learner can link new information with existing knowledge in meaningful ways.

4. Strategic thinking

- The successful learner can create and use a repertoire of thinking and reasoning strategies to achieve complex learning goals.

5. Thinking about thinking

- Higher order strategies for selecting and monitoring mental operations facilitate creative and critical thinking.

6. Context of learning

- Learning is influenced by environmental factors, including culture, technology, and instructional practices.

Motivational and Affective Factors

7. Motivational and emotional influences on learning

- What and how much is learned is influenced by the learner's motivation. Motivation to learn is influenced by the individual's emotional states, beliefs, interests and goals, and habits of thinking.

8. Intrinsic motivation to learn

- The learner's creativity, higher order thinking, and natural curiosity all contribute to motivation to learn. Intrinsic motivation is stimulated by tasks

of optimal novelty and difficulty, relevant to personal interests, and providing for personal choice and control.

9. Effects of motivation on effort

- Acquisition of complex knowledge and skills requires extended learner effort and guided practice. Without learners' motivation to learn, the willingness to exert this effort is unlikely without coercion

Developmental and Social Factors

10. Developmental influences on learning

- As individuals develop, there are different opportunities and constraints for learning. Learning is most effective when differential development within and across physical, intellectual, emotional, and social domains is taken into account

11. Social influences on learning.

- Learning is influenced by social interactions, interpersonal relations, and communication with others.

Individual Differences Factors

12. Individual differences in learning

- Learners have different strategies, approaches, and capabilities for learning that are a function of prior experience and heredity

13. Learning and diversity

- Learning is most effective when differences in learners' linguistic, cultural, and social backgrounds are taken into account

14. Standards and assessment

- Setting appropriately high and challenging standards and assessing the learner as well as learning progress - including diagnostic, process, and outcome assessment - are integral parts of the learning process.

Summary of 14 LCP (Murphy and Alexander)

1. The knowledge base

- One's knowledge serves as the foundation of all future learning.

2. Strategic processing and control

- Learners can develop skills to reflect and regulate their thoughts and behaviors in order to learn more effectively.

3. Motivation and effect

- Factors such as intrinsic motivation, reasons for wanting to learn, personal goals and enjoyment of learning tasks all have a crucial role in the learning process.

4. Development and Individual Differences

- Learning is a unique journey for each person because each learner has his own unique combination of genetic and environmental factors that influence him.

5. Situation or context

- Learning happens in the context of a society as well as within an individual.

HUMAN DEVELOPMENT refers to the physical, cognitive, and psychosocial development of humans throughout the lifespan

DEVELOPMENT - Increase in complexity of function and skill progression

GROWTH – quantifiable amount of change in body dimensions

MATURATION – change that comes with age

LEARNING – change that comes from experiences

Major Principles of Human Development

1. A learner should be considered as a living human being

Factors of Growth:

Heredity

Environment

2. Development is unique to every individual

Elements:

Growth

Structural differences

Alteration in form

3. No two living individuals are exactly the same

DNA, Nipples, Eye Retina, Taste buds, Fingerprint

4. Development is relatively in order

Proximodistal - the tendency for growth to start at the center of the body and work its way outward, toward the extremities

Cephalocaudal – development from head-to-toe progression

5. Children grow in the same pattern

Very rapid growth (First 7 years)

Slowing in the middle (Early to late childhood)

Rapid again (Puberty)

6. Learners respond differently to academic stimuli

7. Almost all human behavior is learned rather than inherited

Approaches to Human Development

TRADITIONAL APPROACH - emphasizes extensive change from birth to adolescence, little or no change in adulthood, and decline in late old age

LIFESPAN APPROACH – developmental change takes place even in adulthood as it does during childhood

Characteristics of Lifespan Development

1. **Lifelong** - development is not completed in infancy or childhood or at any specific age

- it encompasses the entire lifespan, from conception to death
- 2. Plastic** - there are many possible developmental outcomes
- there is no single pathway that must be taken in an individual's development across the lifespan
- 3. Multidimensional** - a complex interplay of factors influence development across the lifespan, including biological, cognitive, and socioemotional changes
- 4. Gradual** - it does not come all on a sudden
- It is also cumulative in nature.
- 5. Contextual** - occurs in context and varies from person to person, depending on factors such as a person's biology, family, school, church, profession, nationality, and ethnicity.
- 6. Involves growth, maintenance, and regulation** - encompasses the broader view and incorporates the important functions of maintenance and regulation
- Rather than seeking growth in various areas, adults seek to maintain performance and/or retard deterioration

DEVELOPMENTAL TASK - a task that arises at or about a certain period in life, unsuccessful achievement of which leads to inability to perform tasks associated with the next period or stage in life

Havighurst's 6 Stages and Characteristics of Development and Developmental Tasks

Infancy and Early Childhood (0-5)

- Learning to walk
- Learning to take solid food
- Learning to talk
- Learning to control the elimination of body wastes
- Getting ready to read

Middle Childhood (6-12)

- Learning to get along w/ age mates
- Learning appropriate masculine or feminine social role
- Developing fundamental skills in learning
- Developing conscience, morality, and a scale of values
- Achieving personal independence
- Developing acceptable attitudes

Adolescence (13-17)

- Establish emotional independence
- Learn skills needed for productive occupation
- Achieve gender-based social role
- Establish mature relationships with peers
- Desiring and achieving socially responsible behavior

Early Adulthood (18-35)

- Selecting a mate
- Learning to live with a marriage partner
- Starting a family; rearing children; managing a home

- Getting started in an occupation

Middle Age (36-60)

- Maintain a standard of living
- Perform civic and social responsibilities
- Maintain a relationship with spouse
- Adjust to physiological changes

Later Maturity / Old Age (Over 60)

- Adjust to deteriorating health
- Adjust to retirement
- Meet social and civil obligations
- Adjust to loss of spouse

JOHN W. SANTROCK'S DEVELOPMENTAL STAGES

PRE-NATAL PERIOD (conception to birth)

Involves tremendous growth – from a single cell to an organism complete with brain and behavioral capabilities

STAGES:

1. Germinal – first 2 weeks
2. Embryonic – 2 - 8 weeks
3. Fetal – 2 - 7 months

TERATOLOGY AND HAZARDS TO PRE-NATAL DEVELOPMENT

Teratology - field of study that investigates the causes of birth defects

Teratogens are any agents from the environment that can cause harm to the developing fetus.

- Many harmful agents cause damage only if exposure occurs during a sensitive period of prenatal development.

Critical factors that influence the degree of harm a teratogen will cause:

- The amount and length of exposure
- Individual differences in susceptibility

Key Teratogens:

1. Prescription and Nonprescription Drugs
2. Psychoactive Drugs (caffeine, alcohol, and nicotine, cocaine, methamphetamine, marijuana, and heroin)
3. Incompatible Blood Types
4. Environmental Hazards (radiation, toxic wastes, chemicals)
5. Maternal Diseases

Risk Factors: Age, Nutrition, Exercise, Stress

INFANCY/BABYHOOD (birth to 2 years old)

Formative years of development

Extreme dependence on adults

Language Development (cry – newborn)

Usually eats every 2 to 3 hours

Uncoordinated movements

Toothless

Poor vision

Responds to human voice and touch

EARLY CHILDHOOD (3 to 5 years old)

Preschool stage

Language and transductive reasoning

Initial socialization experience

Wants to be just like parents

Vivid imagination; indistinct line between real and imaginary

Develops fears

MIDDLE AND LATE CHILDHOOD (6 to 12 y.o.)

School age

Age of conformity

Habit formation stage

Large and small muscles well-developed

Acceptance by peers very important

Parental approval still important

ADOLESCENCE (12 to 18 years old)

Puberty

Period of change

Problem age

Search for identity

Extremely concerned with appearance

EARLY ADULthood/YOUNG ADULT (19-29 yo)

Physical development complete

Emotional maturation continues to develop

Usually learned to accept responsibility for actions and accept criticism

Usually knows how to profit from errors

Socially progress from age-related peer groups to people with similar interests

MIDDLE ADULthood (30 to 60 yo)

Physical changes begin to occur

- Hair begins to thin and gray
- Wrinkles appear
- Hearing and vision decrease
- Muscle lose tone

Main concerns: children, health, job security, aging parents, and fear of aging

Love and acceptance still take a major role

LATE ADULthood (61 years old and above)

Physical deterioration (brittle bones, poor coordination)
Some memory problems
Coping with retirement and forms of entertainment
Very concerned with health and finances
Significant number become depressed; suicide rate is high

9 Temperament Categories

1. the level and extent of motor **activity**
2. the **rhythmicity**, or degree of regularity, of functions such as eating, elimination and the cycle of sleeping and wakefulness
3. the **response** to a new object or person, in terms of whether the child accepts the new experience or withdraws from it
4. the **adaptability** of behavior to changes in the environment
5. the threshold, or **sensitivity** to stimuli
6. the **intensity**, or energy level, of responses
7. the child's general **mood** or "disposition", whether cheerful or given to crying, pleasant or cranky, friendly, or unfriendly
8. the degree of the child's **distractibility** from what he is doing
9. the span of the child's **attention** and his persistence in an activity

Separation anxiety in children is a developmental stage in which the child is anxious when separated from the primary caregiver (usually the mother)

BOWLBY'S ATTACHMENT THEORY

Phases of Attachment

Phase 1: Birth-3 months

- Infants show preference for looking at human faces and listening to human voices
- Respond to people but don't distinguish between them
- 6 weeks: social smiles and eye contact promoting attachment
- Babbling, crying, grasping, and sucking bring infant in closer contact and promotes bonding

Phase 2: 3-6 months

- start to differentiate between people and begin to reserve their attachment behaviors for the people they prefer
- won't do more than stare at a stranger
- If they cry, their favorite people are better able to comfort them
- preferences are restricted to two to three individuals and usually favors one person (anyone who most successfully responded to and had the most positive interactions with the baby)

Phase 3: 6 months to 3 years

- preference for a specific individual becomes more intense: when that individual leaves the room, the infants will have separation anxiety
- attempt to actively follow their favorite person

- after a period of absence, babies will enthusiastically greet them
- At 7 or 8 months old: start to fear strangers (extra caution in the presence of a stranger to crying at the sight of someone new)
- At 1 yo: developed a working model of their favored individual, including how well they respond to the child

Phase 4: 3 years until childhood ends

- at around 3 years old: children start to comprehend that their caretakers have goals and plans of their own
- the child is less concerned when the caretaker leaves for a period of time

Patterns of Attachment

Secure Attachment

- use their caregiver as a secure base from which to explore the world, but if they're frightened or in need of reassurance, they will return
- If caregiver leaves, they will get upset, yet they are confident that their caregiver will return and will greet the caregiver with joy

Avoidant Attachment

- insecure in their attachment to the caregiver
- will not become overly distressed when their caregiver leaves, and upon their return, the child will deliberately avoid the caregiver

Resistant Attachment

- become extremely upset when the parent leaves
- when the caregiver returns their behavior will be inconsistent
 - may initially seem happy to see the caregiver only to become resistant if the caregiver attempts to pick them up
 - often respond angrily to the caregiver; however, they also display moments of avoidance

Disorganized Attachment

- most often displayed by children who have been subject to abuse, neglect, or other inconsistent parenting practices
- seem to be disoriented or confused when their caregiver is present
- seem to view the caregiver as a source of both comfort and fear, leading to disorganized and conflicting behaviors

The Stages of Artistic Development

VIKTOR LOWENFELD'S Stages of Drawing

Stage 1 – Scribble Stage (1 – 3 yo)

- Children are engaged in the physical activity of drawing
- no connection made between the marks and representation
- towards the end of this stage children may begin to give marks names
- mostly about the enjoyment of purely making marks

Stage 2 –Preschematic Stage (3–4 years old)

- beginning to see connections between the shapes that they draw and the physical world around them

- Circles and lines may be described as people or objects that are physically present in the child's life
- A child first makes the connection to communicating through their drawings

Stage 3–The Schematic Stage (5–6 yo)

- assigned shapes to objects that they are attempting to communicate
- developed a schema for creating drawings
- defined order in the development of the drawing
- have a clear separation between the sky and the ground
- Objects are often placed on the ground instead of floating in space
- Objects of importance are often drawn larger than objects of lesser importance

Stage 4–The Dawning Realism (7–9 yo)

- Children are beginning to become more critical of their own work
- While a schema is still used to create drawings, it is more complex than the schema used in earlier stages
- Overlapping can be seen and a sense of spatial relationships is more evident

Stage 5–The Pseudo-Naturalistic Stage

- **(10–13 yo)**
- use of value and light is now apparent in drawings
- Children are very critical of their own success
- Success is determined by the level of realism achieved in the drawing
- Frustration is a common occurrence
- It is exceptionally important to encourage students at this stage

Stage 6 – The Decision Stage (13 – 16 yo)

- Children will decide to continue drawing or view it as an activity without merit
- Because of the level of self-criticism inherent at this stage, many children, (now young adults) view drawing as a skill that they do not possess
- Others decide to continue working on their drawing skills and continue to develop
- Any skill level can be attained with practice
- Most critical stage to the development of an artist

ROBERT SIEGLER'S INFORMATION PROCESSING THEORY

Main Characteristics of IP Approach:

- **Thinking** – information processing
- **Change mechanism** – focus should be on the role of mechanisms of change in development
 - Encoding
 - Automatization
 - Strategy Construction
 - Generalization

Self-modification - self-awareness can enable them to adapt and manage their strategies during problem solving and thinking

RESEARCH in CHILD and ADOLESCENT DEVELOPMENT

Teachers as Consumer/End Users of Research

- Research gives teachers and policy-makers important knowledge to use in decision-making for the benefit of the learners and their families
- Use and integrate the most authoritative research findings
- Enables teachers to come up with informed decision on what to teach and how to teach
- Helps us to be more knowledgeable about how to fit our teaching with the developmental levels of learners

Teachers as Researchers

- The conduct of research does not only belong to thesis and dissertation writers
- Research is for students and teachers

The Scientific Method

- Research is a systematic and logical process
 - **Identify and define the problem**
 - **Determine the hypothesis**
 - **Collect and analyze data**
 - **Formulate Conclusions**
 - **Apply conclusions to the original hypothesis**

Research Designs:

Case Study – an in-depth look at an individual

Correlational Study – determines association

Experimental – determines cause-and-effect relationships

Naturalistic observation – focuses on children's experiences in natural settings

Longitudinal – studies and follows through a single group over a period of time, usually several years or more

Cross-sectional – individuals of different ages are compared at one time

Sequential – combined cross-sectional and longitudinal approaches to learn about lifespan development

Action Research – reflective process of progressive problem-solving led by individuals working with others in teams or as part of a community of practice to improve the way they address issues and solve problems

- stems from their own questions about and reflections on their everyday classroom practice

Data-gathering Techniques:

Observation – can be made in laboratories or natural settings

Physiological Measures – certain indicators of children's development are measured

Standardized – prepared tests that assess individuals' performance in different domains

Interviews and Questionnaires – asking participants to provide information about themselves based on the interview or questionnaires given

Life-History Records – records of information about a lifetime chronology of events and activities

Ethical Principles

1. Research procedures must never harm children, physically or psychologically
2. Children and their families have the right to full information about the research, including possible risks and benefits.
 - Decision must be based on informed consent
3. Children's questions about the research should be answered in a truthful manner and in ways that children can understand
 - Researchers must be honest and clear in their communication
4. There should be respect for privacy.
 - Information obtained through research with children should remain confidential
 - Researchers should not disclose personal information or identity of participants

Data Privacy Act of 2012 (RA 10173)

- To protect the fundamental human right of privacy of communication while ensuring free flow of information to promote innovation and growth
- **The collection of personal data “must be a declared, specified, and legitimate purpose and that...consent is required prior to the collection of all personal data.”**

Impact of Teachers' Research Involvement on Teachers

1. Teachers who have been involved in research may become more reflective, more critical, and analytical in their teaching, and more open and committed to professional development.
2. participating in teacher research also helps teachers become more deliberate in their decision-making and actions in the classroom
3. Teacher research develops professional dispositions of lifelong learning, reflective and mindful teaching, and self-transformation
4. Engaging in teacher research may lead to rethinking and reconstructing what it means to be a teacher or teacher educator and the way teachers relate to children and students
5. Teacher research has potential to demonstrate to teachers and prospective teachers that learning to teach is inherently connected to learning to inquire